

Bylaw 2023

Fall 2025

Experiment M4 Report
Plane Wave Propagation and Polarization

ECE 331 – Electromagnetic Waves (Microwaves Laboratory)

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Objective

The objectives of this experiment are to:

1. Analyze the characteristics of uniform plane waves propagating through different media (lossless and lossy).
2. Observe the attenuation effects in conducting media.
3. Study and visualize different wave polarization states (Linear, Circular, and Elliptical)

Theoretical Background

Wave Propagation A uniform plane wave is characterized by its frequency, wavelength, and phase velocity. These parameters depend on the medium's permittivity, permeability, and conductivity.

. **Lossless Medium (sigma=0):** The wave does not attenuate. The propagation constant is purely imaginary $(k = \omega\sqrt{\mu\epsilon})$ and the intrinsic impedance is real.

Lossy/Conducting Medium $(\sigma \neq 0)$**:** The complex permittivity is used, leading to a complex propagation constant $\gamma = \alpha + j\beta$

- α (Attenuation Constant): Causes exponential decay of the field amplitude ($e^{-\alpha z}$).
- δ_s (Skin Depth): The distance the wave travels before its amplitude decays to $1/e$ of its initial value, defined as $\delta_s = 1/\alpha$
- **Material Classification:** Materials are classified based on the loss tangent $\frac{\sigma}{\omega\epsilon}$.
  - Low loss dielectric: Ratio $\ll 0.01$
  - Quasi-conductor: $0.01 < \text{Ratio} < 100$
  - Good conductor: Ratio $\gg 100$

2.2 Wave Polarization Polarization describes the trace of the electric field vector tip over time at a fixed position.

- **Linear:** x and y components are in phase ($\delta = 0^\circ$ or 180°).
- **Circular:** Components have equal magnitude and a 90° phase difference .
- **Elliptical:** The general case where components differ in magnitude or have a non-standard phase difference
- **Handedness:** Defined by the sign of the phase difference ($\phi_y - \phi_x$). Positive yields Left-Hand (LHP), and negative yields Right-Hand (RHP) .

Part 1: wave propagation

1 Hand Analysis Results:

For a 1 GHz electric field propagating in a non-magnetic medium in the +ve z-direction, with phase = 0, and initial amplitude of 1 V/m, calculate λ, η, α, β, δs, the amplitude of the electric field after it travels 1 λ, and the type of material for each of the following propagation mediums and place your results in a neat table:

a) Vacuum: εr = 1, σ = 0

b) Fresh water: εr = 81, σ = 0

c) Copper: εr = 1, σ = 5.8*107 S/m

d) Dry soil: εr = 2.5, σ = 10-4 S/m

Hand analysis :-

$f = 10^9$ Hz $\mu_r = 1$ $\vec{E} = +\hat{z}$ $E_0 = 1 \left(\frac{V}{m}\right)$

A Vacuum $\varepsilon_r = 1$ $\sigma = 0$, $\varepsilon'' = 0$ ∴ loss less medium

$\beta = K = \omega\sqrt{\mu\varepsilon} = 20,96 \left(\frac{rad}{m}\right)$, $\lambda = \frac{2\pi}{K} = 0,3$ (m) , $\alpha = 0$

$\delta_s = \frac{1}{\alpha} = \infty$, $\eta_0 = \sqrt{\mu_0/\varepsilon_0} = 377 (\Omega)$ $\|\vec{E}\| = 1 \left(\frac{V}{m}\right)$ after 1λ

B Fresh Water $\varepsilon_r = 81$, $\alpha = 0$ ∴ loss less dielectric

$\beta = K = \omega\sqrt{\mu\varepsilon} = 188,63 \left(\frac{rad}{m}\right)$, $\lambda = \frac{2\pi}{K} = 0,033$ (m) , $\alpha = 0$

$\delta_s = \frac{1}{\alpha} = \infty$, $\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = 41,89 (\Omega)$ $\|\vec{E}\| = 1 \left(\frac{V}{m}\right)$ after 1λ

C Copper $\varepsilon_r = 1$, $\sigma = 5,8 \times 10^7 \left(\frac{S}{m}\right)$, $\frac{\sigma}{\omega\varepsilon} \simeq 10^9 \gg 100$ ∴ good Conductor

$\gamma = (1+j)\sqrt{\omega\mu\sigma/2} = [8,51 + 478,51 j] \times 10^3$

∴ $\beta = 478,51 \times 10^3 \left(\frac{rad}{m}\right)$, $\alpha = 478,51 \times 10^3 \left(\frac{Np}{m}\right)$, $\lambda = \frac{2\pi}{\beta} = 1,31 \times 10^{-5}$ (m)

$\delta_s = \frac{1}{\alpha} = 2,1 \times 10^{-6}$ (m) , $\eta = \frac{1+j}{\sigma\delta_s} = [8,21 + 8,21 j] \times 10^{-3} (\Omega)$

$\|\vec{E}\| = E_0 e^{-\alpha\lambda} = 1,855 \times 10^{-3} \left(\frac{V}{m}\right)$ after 1λ

D Dry Soil $\varepsilon_r = 2,5$, $\sigma = 10^{-4} \left(\frac{S}{m}\right)$ $\frac{\sigma}{\omega\varepsilon} = 7,2 \times 10^{-4} \ll \frac{1}{100}$ ∴ low loss dielectric

$\gamma = j\omega\sqrt{\mu\varepsilon}\left(1 - \frac{j}{2}\frac{\sigma}{\omega\varepsilon}\right) = 33,14 j + 0,012$

∴ $\beta = 33,14 \left(\frac{rad}{m}\right)$, $\alpha = 0,012 \left(\frac{Np}{m}\right)$, $\lambda = \frac{2\pi}{\beta} = 0,19$ (m)

$\delta_s = \frac{1}{\alpha} = 83,33$ (m) $\eta = \frac{j\omega\mu}{\gamma} = 238,25 + 0,086 j (\Omega)$

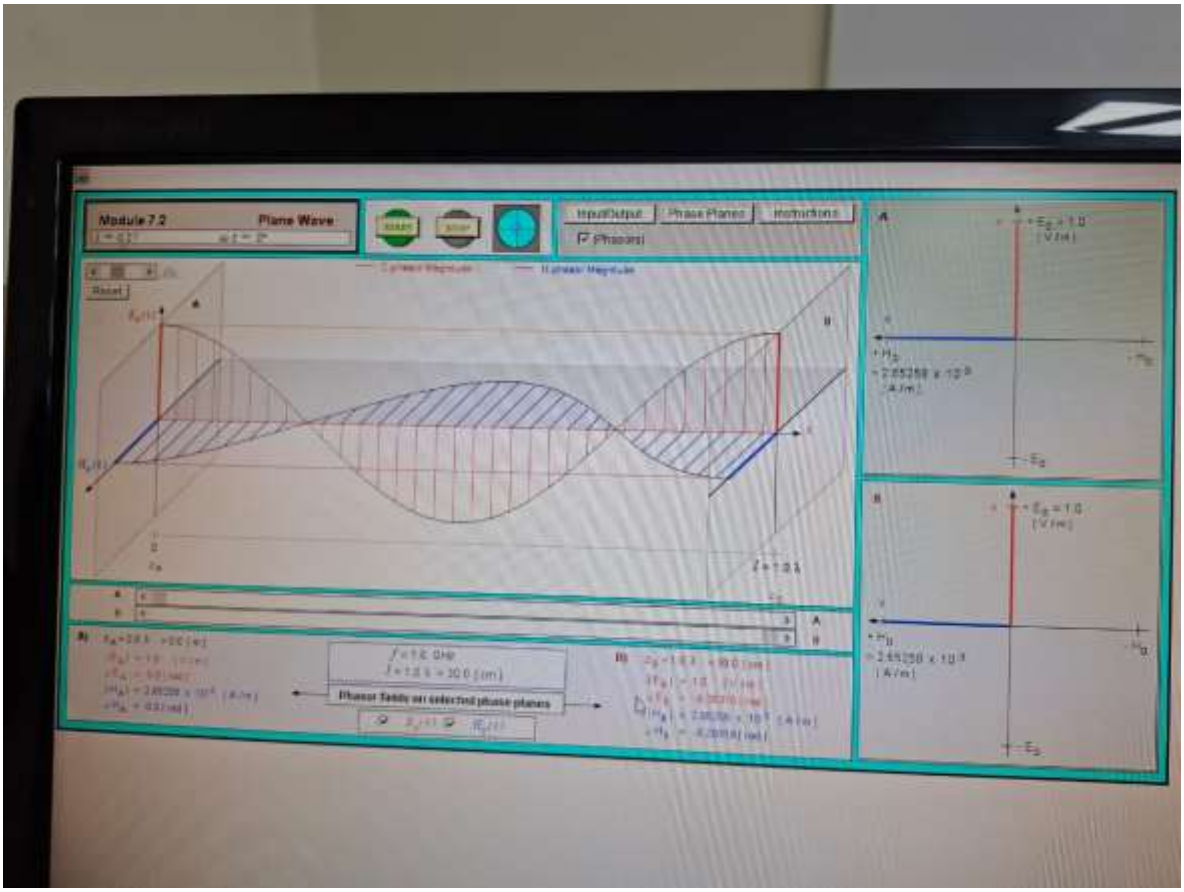
$\|\vec{E}\| = E_0 e^{-\alpha\lambda} = 0,998 \left(\frac{V}{m}\right)$

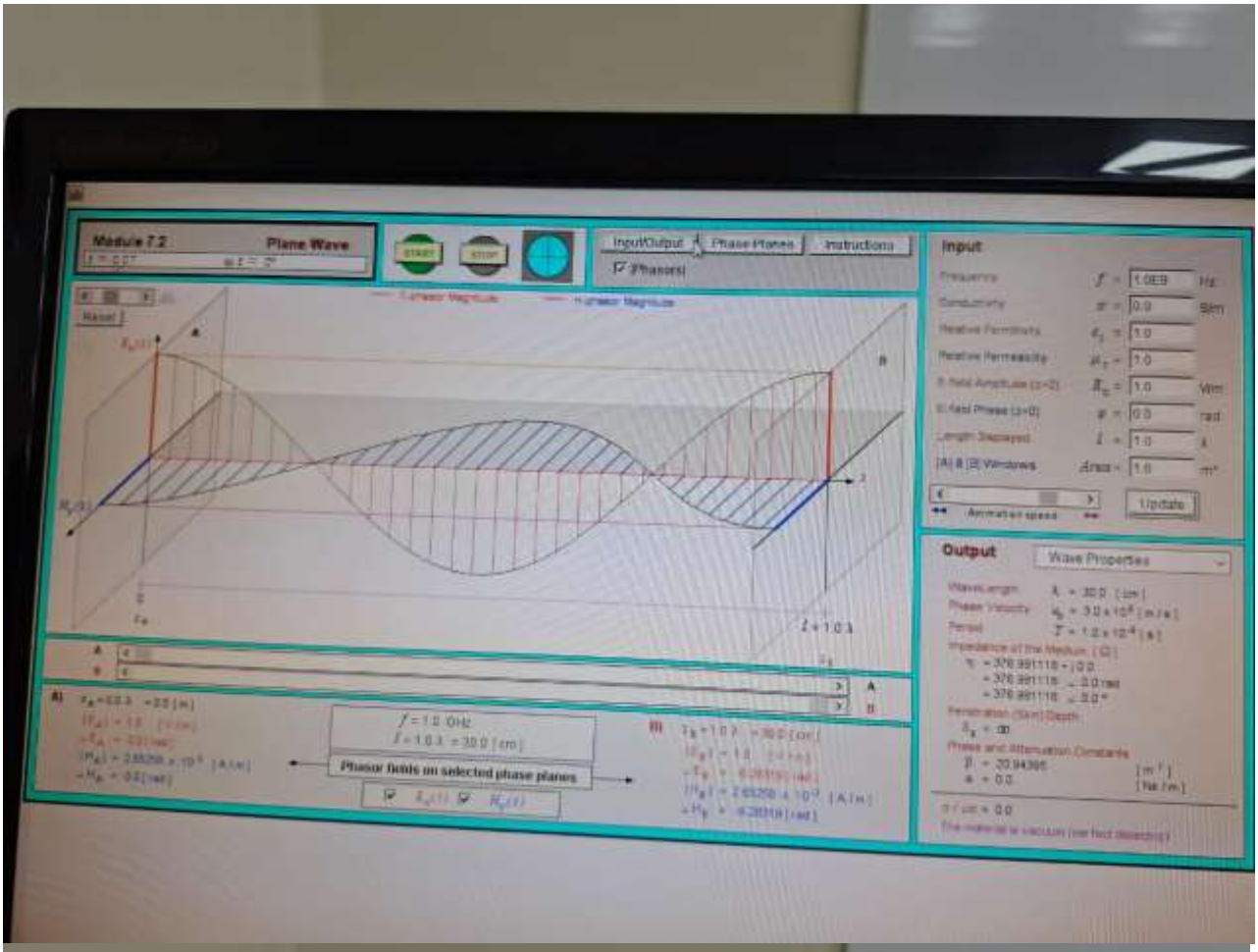
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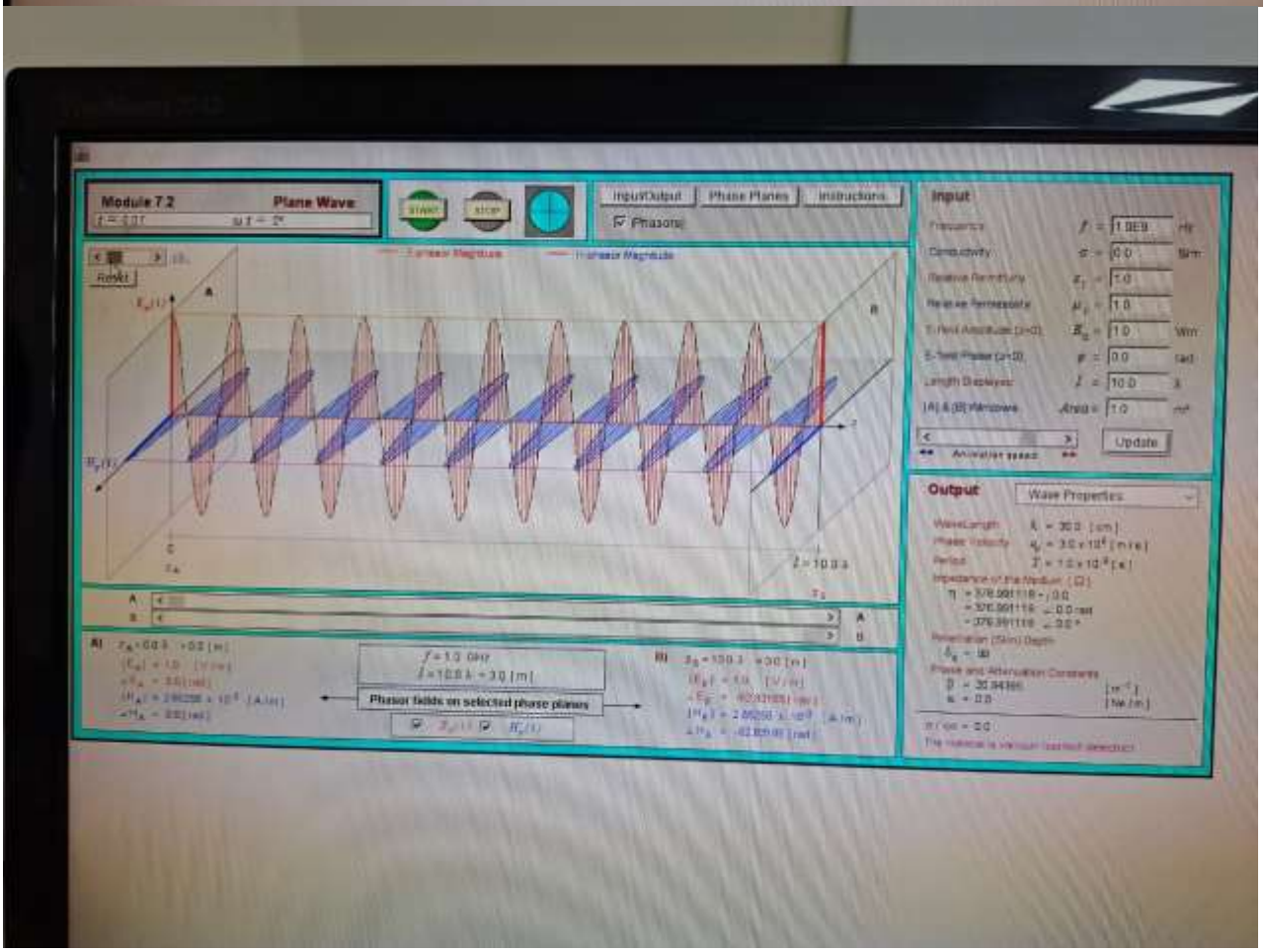
Medium	Type	λ(m)	α ($\frac{NP}{m}$)	β ($\frac{rad}{m}$)	δ_s(m)	η (Ω)
Vacuum	lossless dielectric	0,3	Zero	20,96	∞	377
Fresh water	lossless dielectric	0,033	Zero	188,6	∞	41,89
Copper	good Conductor	$1,31 \times 10^{-3}$	$4,76 \times 10^{3}$	$4,78,5 \times 10^{3}$	$2,1 \times 10^{-6}$	$[8,21 + 8,21 J] \times 10^{-3}$
Dry Soil	low loss dielectric	0,19	0,012	33,14	83,33	238,25 + 0,086 J

2 Simulation Results The following figures show the standing wave patterns generated

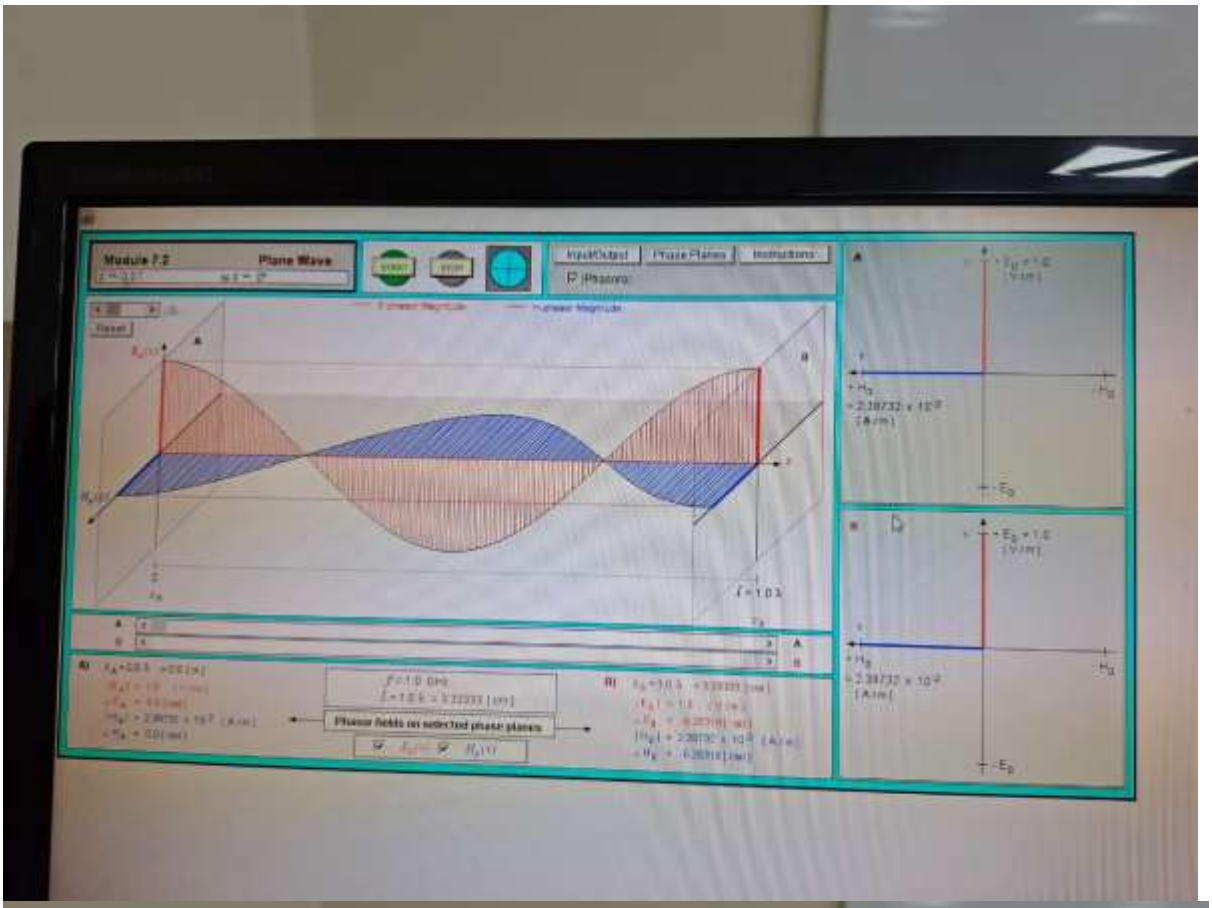
Case (i): Vacuum: $\varepsilon r = 1$, $\sigma = 0$

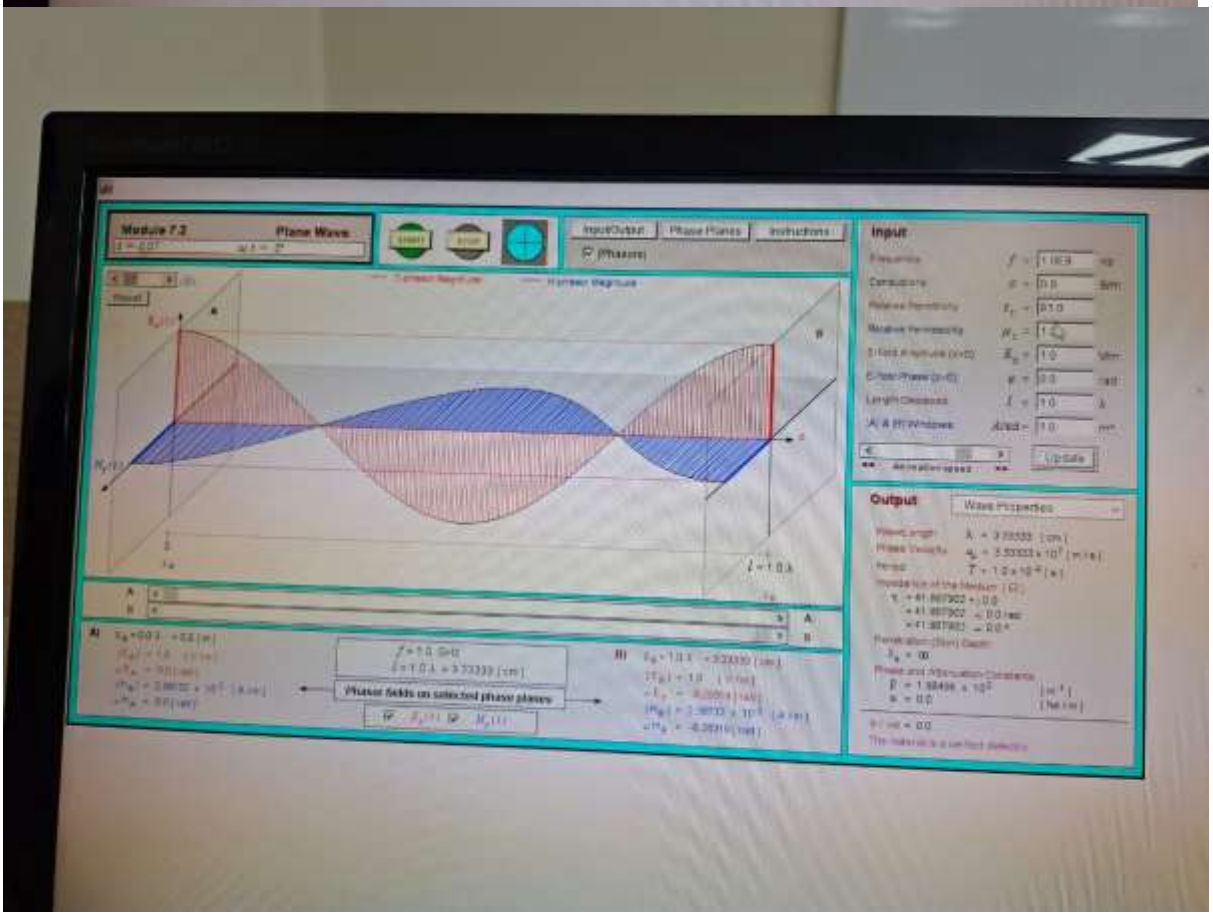


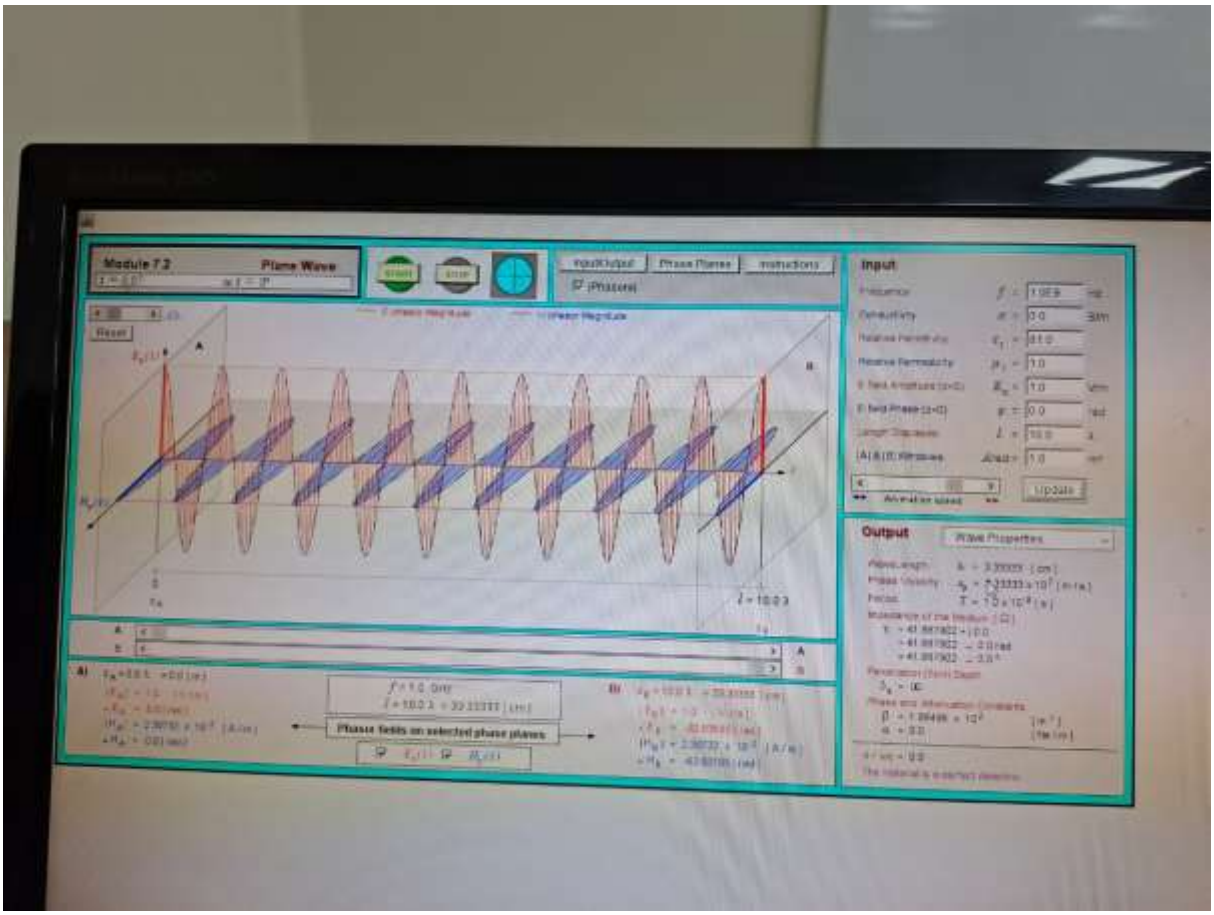

Module 7.2
Plane Wave
START
STOP
Input/Output
Phase Planes
Instructions
Phasors
Input
Update
Output
Wave Properties
Phasor fields on selected phase planes


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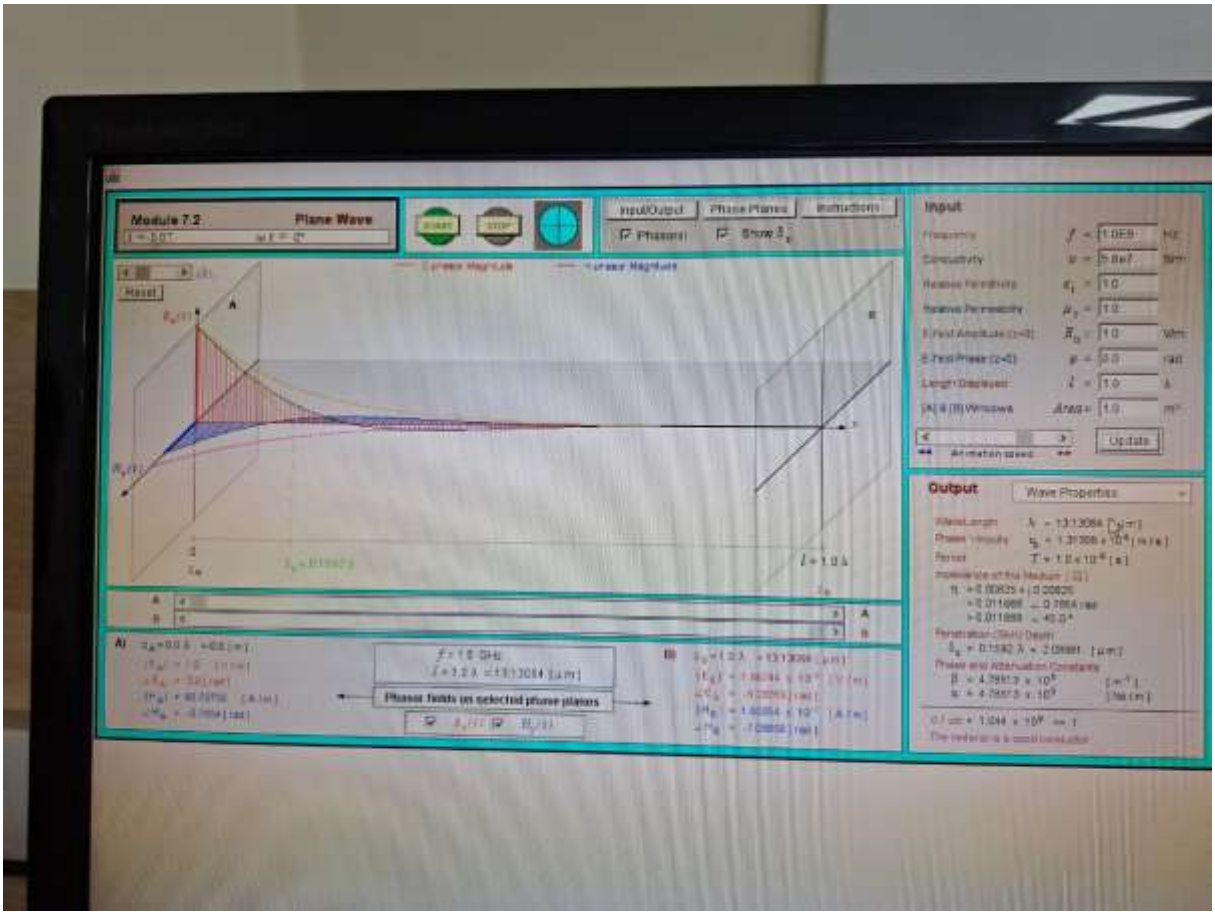
Case (ii): Fresh water: $\varepsilon r = 81$, $\sigma = 0$

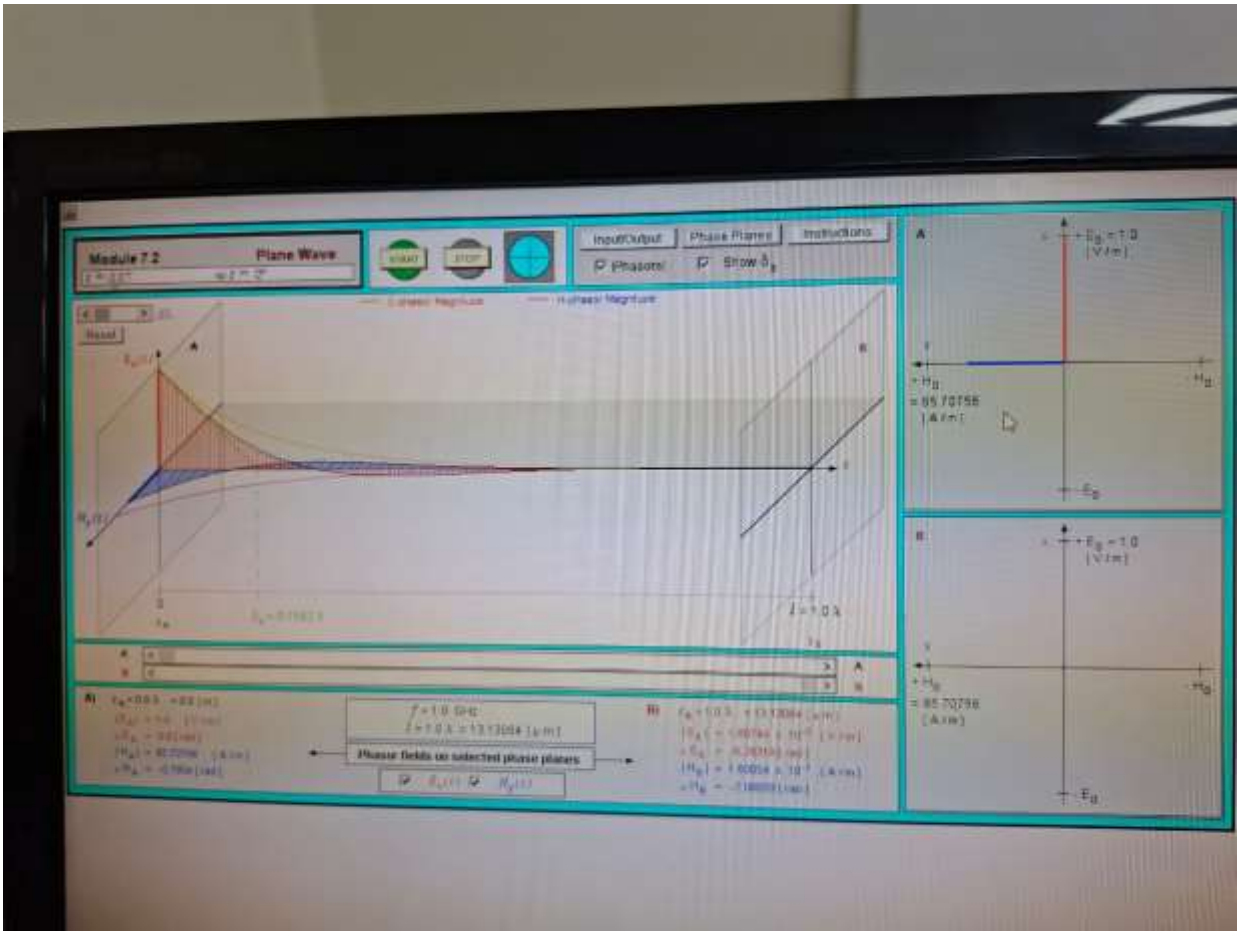




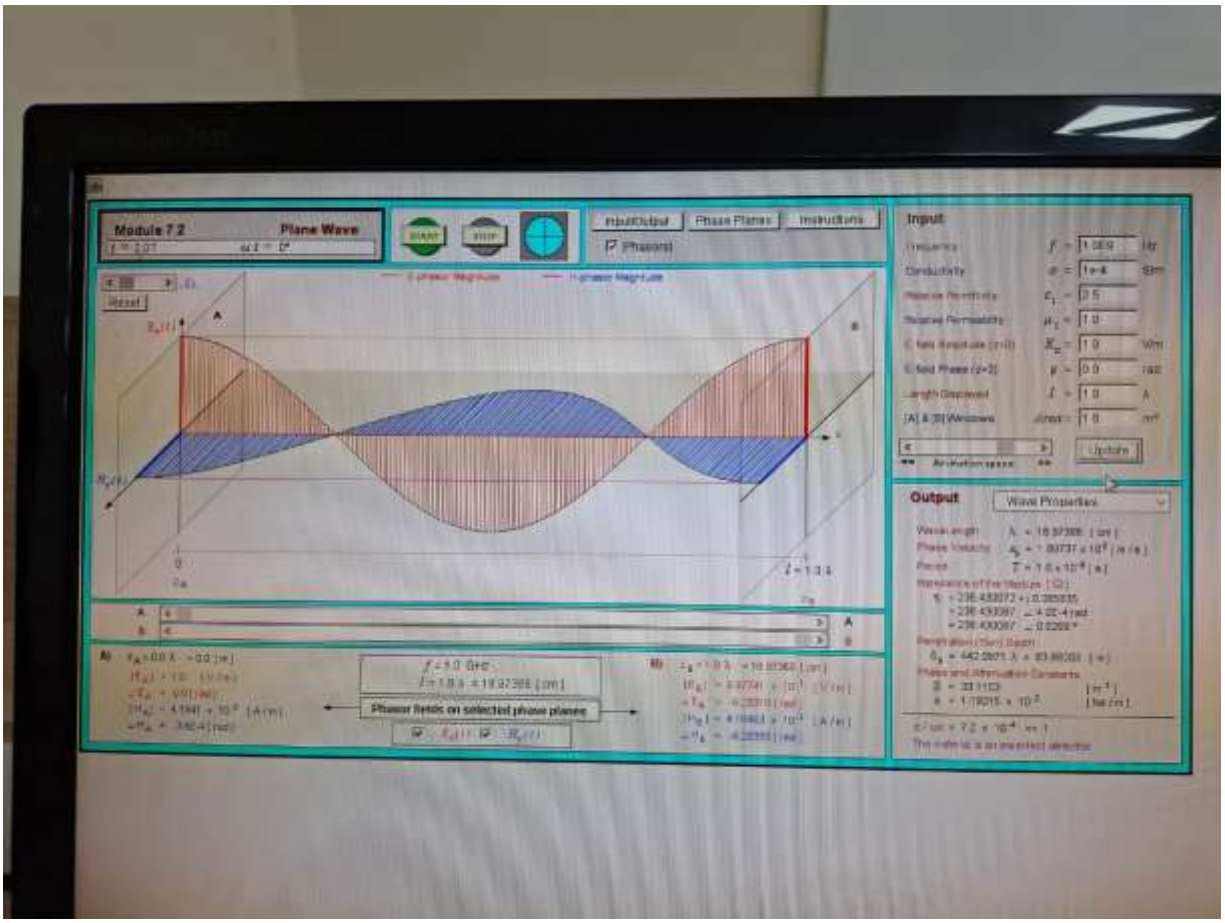


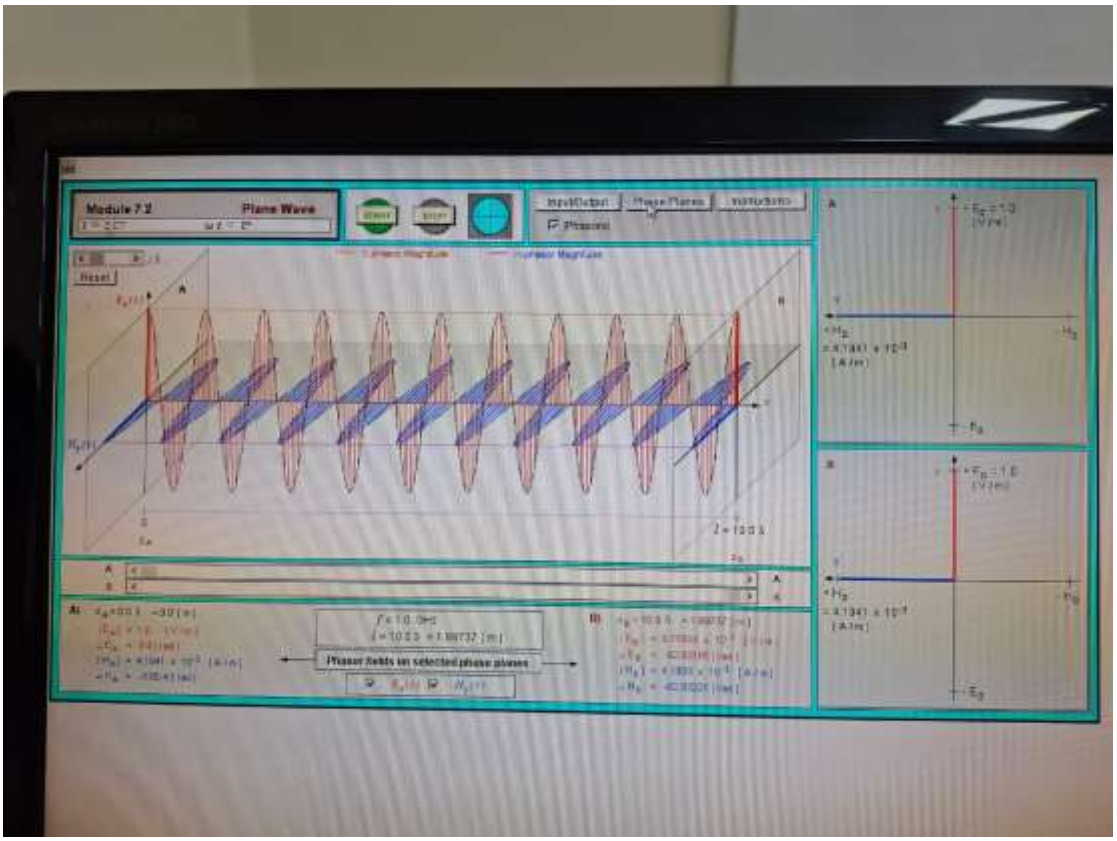
Case (iii): Copper: εr = 1, σ = 5.8*107 S/m





Case (iv): Dry soil: εr = 2.5, σ = 10-4 S/m





Part 2: Wave polarization

1 Hand Analysis Results:

For a wave characterized by the electric field

$$E(z,t) = \hat{x}\, ax\, cos(\omega t - kz) + \hat{y}\, ay\, cos(\omega t - kz + \delta)$$

Identify the polarization state, and sketch the locus of E(0, t) for each of the following cases:

(a) ax = 0.3 V/m, ay = 0.4 V/m, and δ = 0◦

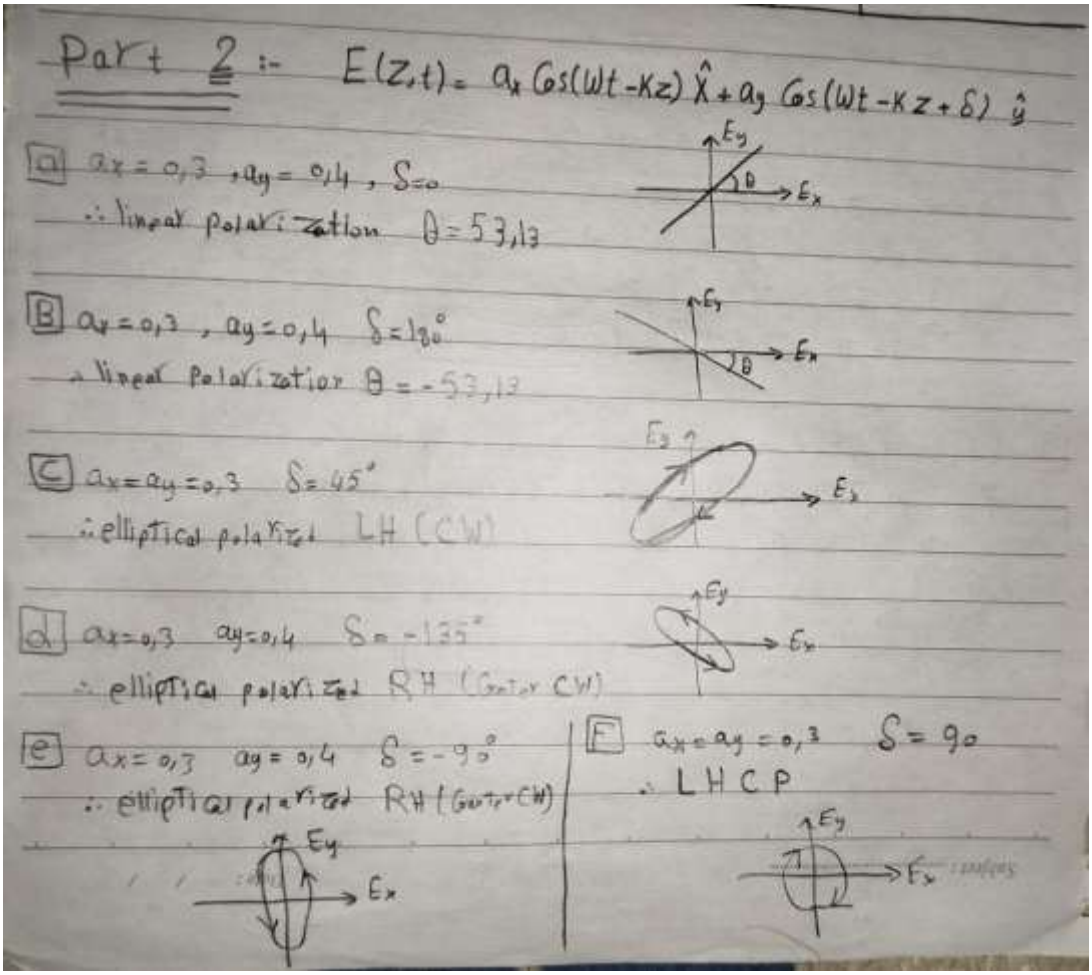
(b) ax = 0.3 V/m, ay = 0.4 V/m, and δ = 180◦

(c) ax = 0.3 V/m, ay = 0.3 V/m, and δ = 45◦

(d) ax = 0.3 V/m, ay = 0.4 V/m, and δ = −135◦

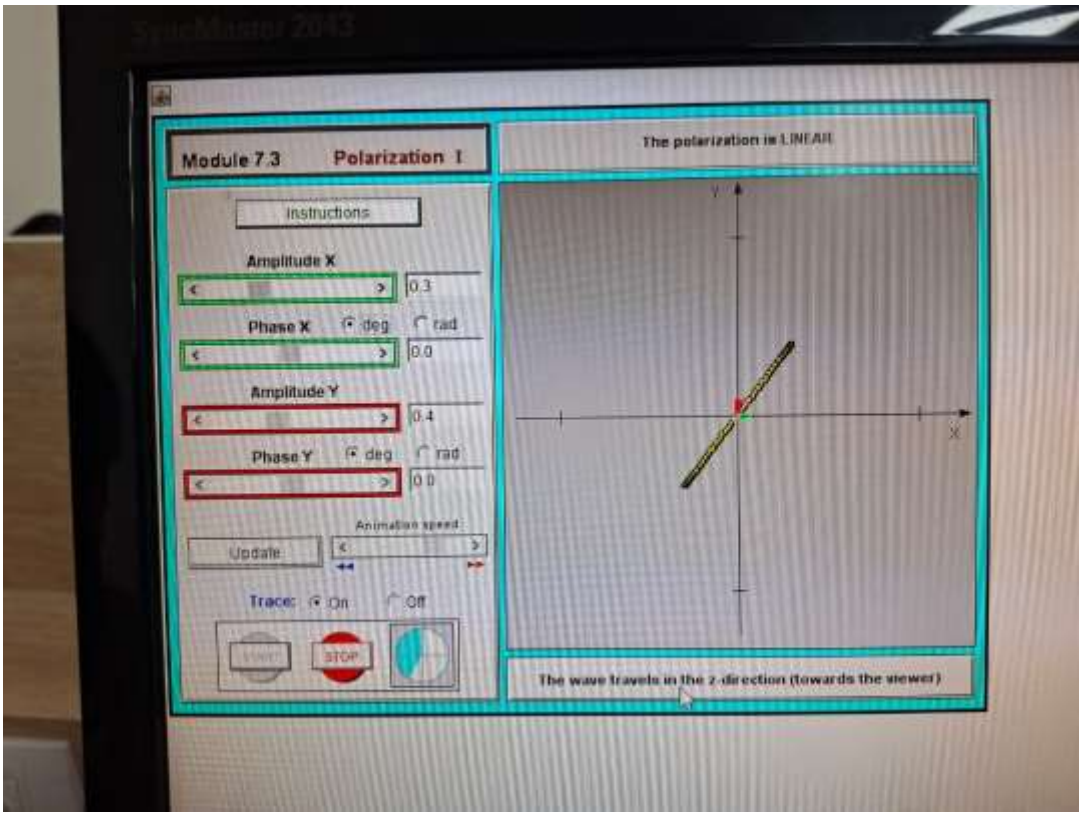
(e) ax = 0.3 V/m, ay = 0.4 V/m, and δ = −90◦

(f) ax = 0.3 V/m, ay = 0.3 V/m, and δ = 90◦

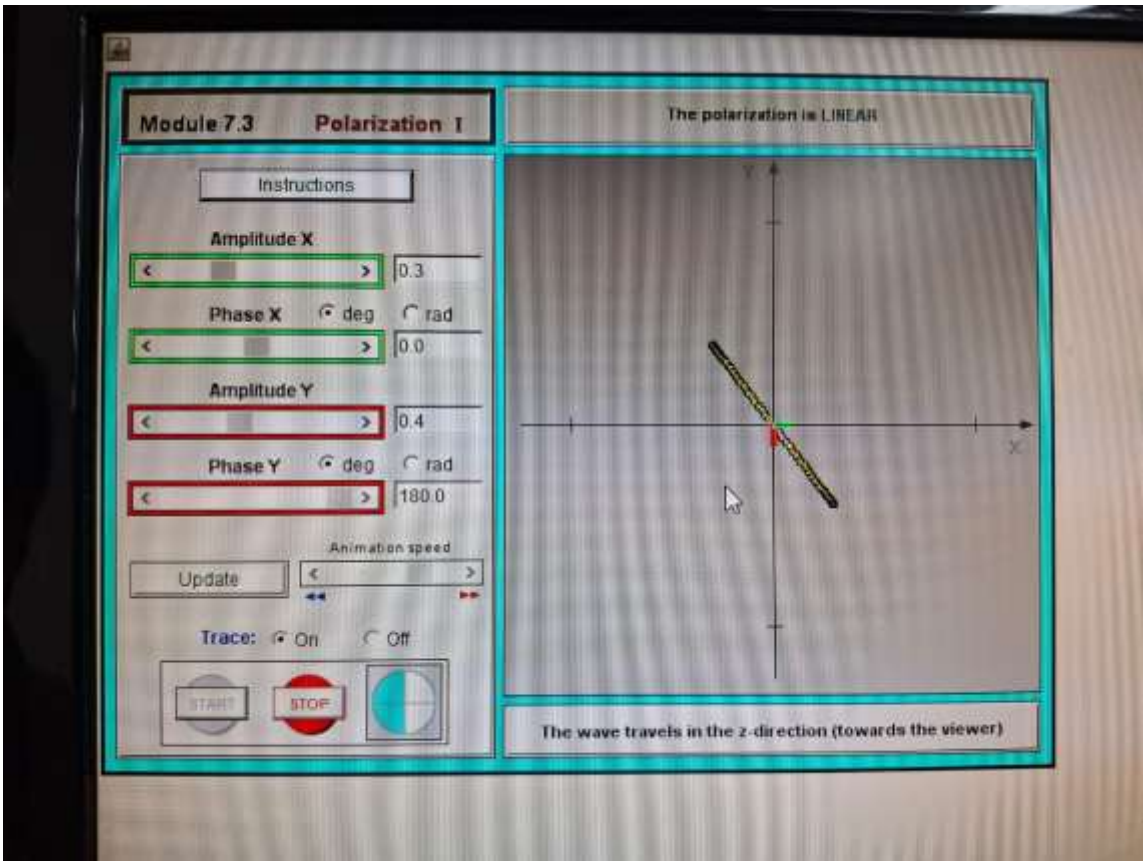


2 Simulation Results

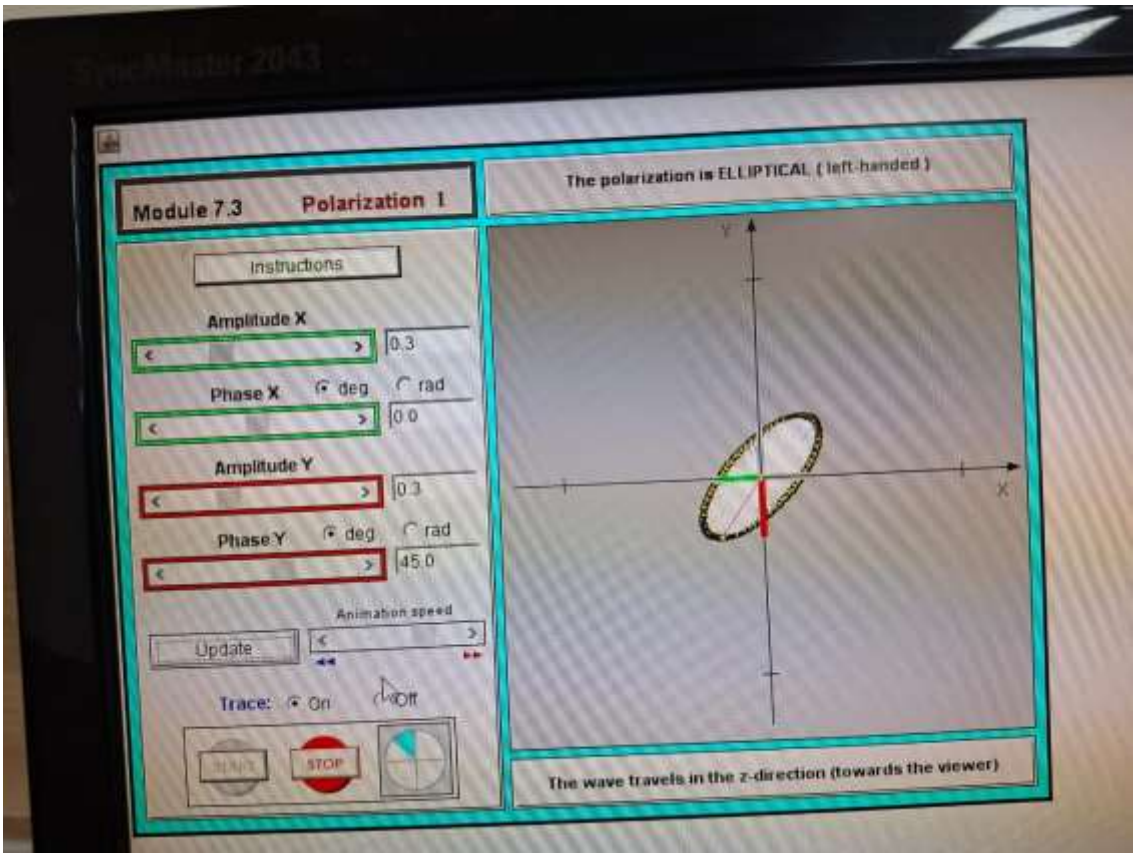
Case (i): ax = 0.3 V/m, ay = 0.4 V/m, and δ = 0◦



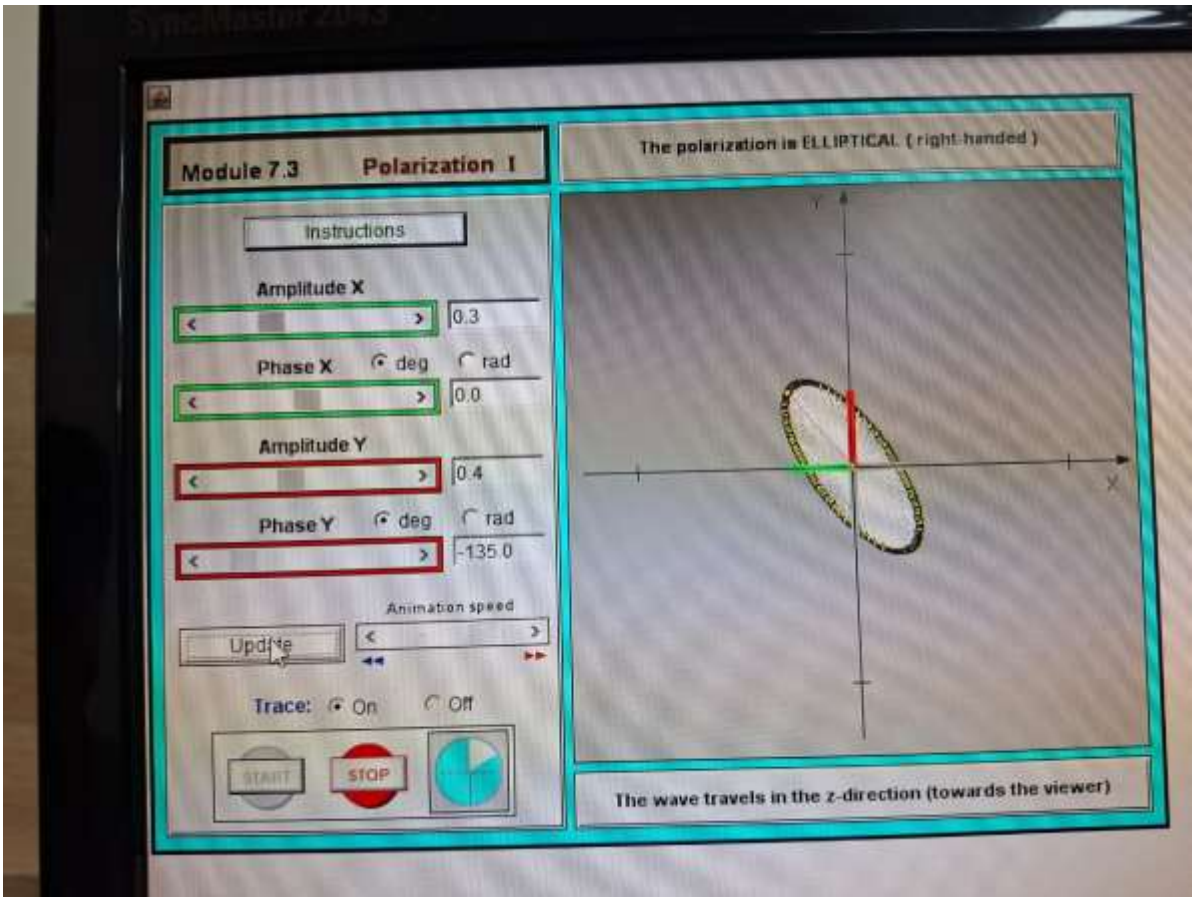
Case (ii): ax = 0.3 V/m, ay = 0.4 V/m, and δ = 180◦



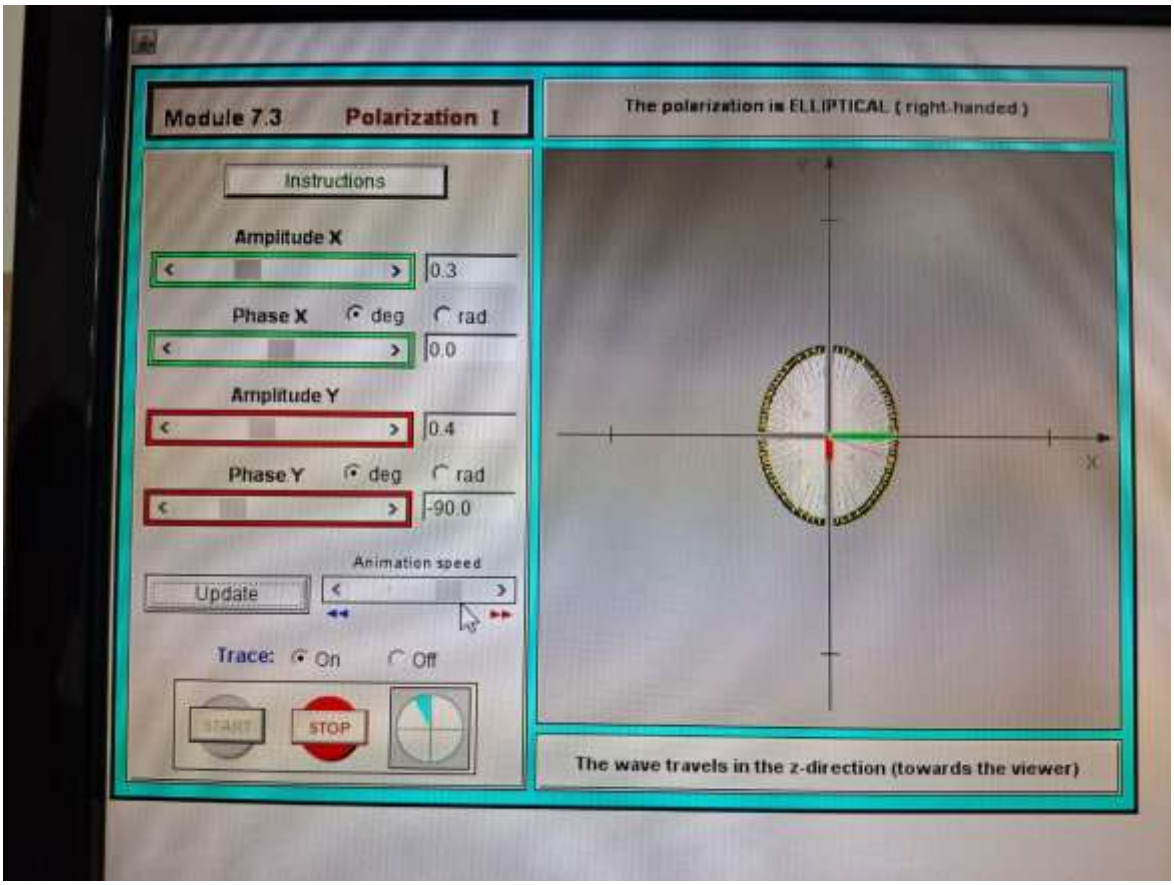
Case (iii): ax = 0.3 V/m, ay = 0.3 V/m, and δ = 45◦



Case (iv): ax = 0.3 V/m, ay = 0.4 V/m, and δ = −135◦



Case (v): ax = 0.3 V/m, ay = 0.4 V/m, and δ = −90◦



Case (vi): ax = 0.3 V/m, ay = 0.3 V/m, and δ = 90◦

